Mississippi Valley State University

CS 341 — Discrete Structures

Lecture 3: When Are Two Statements the Same?

Logical Equivalence and the Laws of Logic

Chartrand & Zhang, Section 1.2, pp. 18–24

Name: _____ Date: _____

Where we are

Wednesday you built a truth table and discovered that $\sim(P \wedge Q)$ and $(\sim P) \wedge (\sim Q)$ are different, while $\sim(P \wedge Q)$ and $(\sim P) \vee (\sim Q)$ are the same. Today we say precisely what "the same" means, prove it, and collect the handful of laws that let you rewrite any compound statement without changing what it says.

By the end of today you should be able to

1. State what it means for two compound statements to be logically equivalent.
2. Prove an equivalence with a truth table — and disprove one with a single row.
3. State and apply De Morgan's Laws.
4. Use the commutative, associative, and distributive laws to rewrite a statement.
5. Explain why logical equivalence is not the same as interchangeability in code.

Warm-up (no looking back)

Rewrite each without an outer negation.

(a) $\sim(P \vee Q) \quad \equiv \quad \underline{(\sim P) \wedge (\sim Q)}$

(b) $\sim(P \wedge Q) \quad \equiv \quad \underline{(\sim P) \vee (\sim Q)}$

Both have names. Today we find out what they are.

1. Logical equivalence

Definition 1.21 — logical equivalence

Two compound statements R and S built from the same component statements are **logically equivalent** if they have the *same truth value for every possible combination* of truth values of their components. We write $R \equiv S$.

Two jobs, two amounts of work

To prove $R \equiv S$: build both columns and check that they agree in every row. Nothing may be skipped.
To prove $R \not\equiv S$: exhibit one row where they differ. One is enough.
That asymmetry should look familiar. Establishing a general claim requires covering every case; refuting one requires a single counterexample. The only reason we can finish the job here is that "every case" means 2^n rows — a finite number we can actually exhaust.

2. Commutative Laws

Theorem 1.22 — Commutative Laws

For statements P and Q:

$$P \wedge Q \equiv Q \wedge P \qquad P \vee Q \equiv Q \vee P$$

P	Q	$P \wedge Q$	$Q \wedge P$	$P \vee Q$	$Q \vee P$
T	T	T	T	T	T
T	F	F	F	T	T
F	T	F	F	T	T
F	F	F	F	F	F

Logically equivalent is not the same as interchangeable in code

These two lines are logically equivalent. They are *not* interchangeable:

```
if p is not None and p.x > 0: # safe
if p.x > 0 and p is not None: # crashes when p is None
```

The reason is that `and` in a program does two jobs. It computes a truth value, and it also decides *whether the second expression gets evaluated at all.* Logic only describes the first job. When you reorder a condition in real code, you are changing something the truth table cannot see.
Keep both facts. The equivalence is real, and so is the crash.

3. De Morgan's Laws

Theorem 1.24 — De Morgan's Laws

For statements P and Q:

$$\sim(P \vee Q) \equiv (\sim P) \wedge (\sim Q) \qquad \sim(P \wedge Q) \equiv (\sim P) \vee (\sim Q)$$

P	Q	$\sim(P \wedge Q)$	$(\sim P) \vee (\sim Q)$	$\sim(P \vee Q)$	$(\sim P) \wedge (\sim Q)$
T	T	F	F	F	F
T	F	T	T	F	F
F	T	T	T	F	F
F	F	T	T	T	T

That table is the whole proof. Every row agrees, and there are no other rows.

How to say them in English.

- "Not (P or Q)" means neither P nor Q.
- "Not (P and Q)" means at least one of them fails.

Apply them. Write each negation with no outer negation.

Statement	Negation
$x > 0$ and $y > 0$	$x \leq 0$ or $y \leq 0$
The connection is secure or the data is encrypted.	The connection is not secure and the data is not encrypted.
The integers a and b are both nonnegative.	a is negative or b is negative.

4. Double Negation

Theorem 1.28 — Double Negation

For every statement P: $P \equiv \sim(\sim P)$

P	$\sim P$	$\sim(\sim P)$
T	F	T
F	T	F

Small, but you will use it constantly — usually to clean up after applying De Morgan.

5. Associative and Distributive Laws

Theorem 1.31 — Associative and Distributive Laws

For statements P, Q, and R:

$$P \vee (Q \vee R) \equiv (P \vee Q) \vee R \qquad P \wedge (Q \wedge R) \equiv (P \wedge Q) \wedge R$$
$$P \vee (Q \wedge R) \equiv (P \vee Q) \wedge (P \vee R) \qquad P \wedge (Q \vee R) \equiv (P \wedge Q) \vee (P \wedge R)$$

Let us prove the last one.

P	Q	R	$Q \vee R$	$P \wedge (Q \vee R)$	$P \wedge Q$	$P \wedge R$
T	T	T	T	T	T	T
T	T	F	T	T	T	F
T	F	T	T	T	F	T
T	F	F	F	F	F	F
F	T	T	T	F	F	F
F	T	F	T	F	F	F
F	F	T	T	F	F	F
F	F	F	F	F	F	F

Now read off $(P \wedge Q) \vee (P \wedge R)$ from the last two columns and compare it with the fifth. Do they agree in all eight rows? Yes

Associativity does not let you drop parentheses across different connectives

$P \wedge Q \wedge R$ is unambiguous — associativity says the grouping does not matter.
But $P \wedge (Q \vee R)$ and $(P \wedge Q) \vee R$ are **different statements**. Take P false, Q false, R true: the first is F and the second is T. When $\wedge$ and $\vee$ are mixed, the parentheses are carrying meaning.

A useful analogy, and where it breaks

Think of $\wedge$ as multiplication and $\vee$ as addition. Then $P \wedge (Q \vee R) \equiv (P \wedge Q) \vee (P \wedge R)$ looks exactly like $a(b+c) = ab + ac$, which is true in arithmetic.
Now try the other one. Is $a + (b \times c) = (a+b) \times (a+c)$? Take $a = 1$, $b = 2$, $c = 3$: the left side is 7 and the right side is 12. In arithmetic the second distributive law is **false**.
So: does the second distributive law hold in logic? The theorem box above claims it does — but a claim is not a proof, and here the analogy actively predicts the wrong answer. **Homework 2, Problem 3 asks you to settle it with a truth table.** Do not assume.

6. Putting it together — and Wednesday's promise

On Wednesday you built the table for $(\sim P) \vee Q$ and were told to keep the page. Here is why.

Start with a statement that means *"it never happens that P holds while Q fails":*

$$\sim(P \wedge (\sim Q))$$

Simplify it, one law at a time:

Expression	**Law used**
$\sim(P \wedge (\sim Q))$	(start)
$\equiv (\sim P) \vee (\sim(\sim Q))$	De Morgan (Thm 1.24)
$\equiv (\sim P) \vee Q$	Double negation (Thm 1.28)

So "you can never have P without Q" and $(\sim P) \vee Q$ are the *same statement.* Check it against Wednesday's table: false in exactly one row, when P is true and Q is false.

Monday

That compound statement is important enough to get its own name and its own symbol. It is called the **implication**, written $P \Rightarrow Q$ and read "if P, then Q" — Section 1.3, which begins on p. 25. Almost every theorem you will meet for the rest of this course is an implication. You have now met it twice before being introduced.

The laws — keep this page

Commutative (Thm 1.22)	$P \wedge Q \equiv Q \wedge P$ $\quad$ $P \vee Q \equiv Q \vee P$
De Morgan (Thm 1.24)	$\sim(P \vee Q) \equiv (\sim P) \wedge (\sim Q)$ $\sim(P \wedge Q) \equiv (\sim P) \vee (\sim Q)$
Double negation (Thm 1.28)	$P \equiv \sim(\sim P)$
Associative (Thm 1.31)	$P \vee (Q \vee R) \equiv (P \vee Q) \vee R$ $P \wedge (Q \wedge R) \equiv (P \wedge Q) \wedge R$
Distributive (Thm 1.31)	$P \vee (Q \wedge R) \equiv (P \vee Q) \wedge (P \vee R)$ $P \wedge (Q \vee R) \equiv (P \wedge Q) \vee (P \wedge R)$

Exit ticket

1. Write $\sim(A \vee B)$ without an outer negation, and name the law you used.

2. True or false: $P \wedge (Q \vee R) \equiv (P \wedge Q) \vee R$. Justify in one line.

3. What must you show to *prove* two compound statements are equivalent? What must you show to *disprove* it?

Before Monday

Homework 1 is due tonight — Monday, August 24, 11:59 p.m.

One PDF to Canvas, with the completed collaboration and AI statement as the first item.

Read Section 1.3, "Implications," beginning on p. 25. Read at least through Definition 1.33 and the truth table in Figure 1.9.

Self-check (not collected). Section 1.2, Exercises 13, 16(a), 16(b), and 17. Exercise 17 asks you to verify $P \vee (P \wedge Q) \equiv P$ — a result we did not cover, and a satisfying one to find on your own.

Homework 2 has also been posted and due Saturday, August 29.

Questions I still have